Problem Set 75

Related Rates

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Problem 1

If V is the volume of a cube with edge length x and the cube expands as time passes, find $\frac{dV}{dt}$ in terms of $\frac{dx}{dt}$.

Given: x

Unknown: $\frac{dV}{dt}$

$$V = x^3$$
$$\frac{dV}{dt} = \boxed{3x^2\frac{dx}{dt}}$$

Problem 2

Each side of a square is increasing at a rate of 6 cm/s. At what rate is the area of the square increasing when the area of the square is 16 cm^2?

Given: $\frac{ds}{dt} = 6$ cm/s

Unknown: $\left.\frac{dA}{dt}\right|_{A=16}$

$$A = s^2$$
$$\frac{dA}{dt} = 2s \cdot \frac{ds}{dt} = 2 \cdot 4 \cdot 6 = \boxed{48\ \text{cm}^2/\text{s}}$$

The area of the square is increasing at a rate of 48 cm^2/s.

Problem 4

Suppose $y = x^3 + 2x$, where x and y are functions of t. If $\frac{dx}{dt} = 5$, find $\frac{dy}{dt}$ when $x = 2$.

Given: $y = x^3 + 2x$, $\frac{dx}{dt} = 5$

Unknown: $\left.\frac{dy}{dt}\right|_{x=2}$

$$\frac{dy}{dt} = 3x^2 \cdot \frac{dx}{dt} + 2 \cdot \frac{dx}{dt} = 3 \cdot 4 \cdot 5 + 2 \cdot 5 = \boxed{70}$$

Problem 6

A plane flying horizontally at an altitude of 1 mi and a speed of 500 mi/h passes directly over a radar station. Find the rate at which the distance from the plane to the station is increasing when it is 2 mi away from the station.

Given: $h = 1$ mi, $\dfrac{dx}{dt} = 500$ mi/h

Unknown: $\left.\dfrac{dy}{dt}\right|_{x=2}$

$$y^2 = h^2 + x^2 = 1 + x^2$$
$$2y \cdot \frac{dy}{dt} = 2x \cdot \frac{dx}{dt}$$
$$\frac{dy}{dt} = \frac{x}{\sqrt{1+x^2}} \cdot \frac{dx}{dt} = \frac{2}{\sqrt{5}} \cdot 500 = \boxed{200\sqrt{5}\text{ mi/h}}$$

The distance from the plane to the station is increasing at a rate of $200\sqrt{5}$ mi/h.

Problem 8

Two cars start moving from the same point. One travels south at 60mi/h and the other travels west at 25mi/h. At what rate is the distance between the cars increasing two hours later?

Given: $\dfrac{dx}{dt} = 60$ mi/h, $\dfrac{dy}{dt} = 25$ mi/h

Unknown: $\left.\dfrac{dz}{dt}\right|_{t=2\text{h}}$

$$z^2 = x^2 + y^2$$
$$x = \frac{dx}{dt} \cdot t = 120mi$$
$$y = \frac{dy}{dt} \cdot t = 50mi$$
$$2z \cdot \frac{dz}{dt} = 2x \cdot \frac{dx}{dt} + 2y \cdot \frac{dy}{dt}$$
$$\frac{dz}{dt} = \frac{x \cdot \frac{dx}{dt} + y \cdot \frac{dy}{dt}}{\sqrt{x^2+y^2}} = \frac{120 \cdot 60 + 25 \cdot 50}{130} = \boxed{65\text{ mi/h}}$$

The distance between the cars increases at a rate of 65 mi/h.

Problem 10

The altitude of a triangle is increasing at a rate of 1 cm/min while the area of the triangle is increasing at a rate of 2 cm^2/min. At what rate is the base of the triangle changing when the altitude is 10 cm and the area is 100 cm^2?

Given: $\dfrac{dh}{dt} = 1$ cm/min, $\dfrac{dA}{dt} = 2$ cm^2/min.

Unknown: $\left.\dfrac{dB}{dt}\right|_{h=10,A=100}$

$$B = \frac{2A}{h}$$

$$\frac{dB}{dt} = \frac{2(h \cdot \frac{dA}{dt} - A \cdot \frac{dh}{dt})}{h^2} = \frac{2(10 \cdot 2 - 100 \cdot 1)}{100} = \boxed{-1.6 \text{ cm/min}}$$

The base of the triangle changes at a rate of -1.6 cm/min.

Problem 12

Water is leaking out of an inverted conical tank at a rate of 10,000 cm^3/min at the same time that water is being pumped into the tank at a constant rate. The tank has height 6 m and the diameter at the top is 4 m. If the water level is rising at a rate of 20 cm/min when the height of the water is 2m, find the rate at which water is being pumped into the tank.

Given: $h = 6\text{m}$, $\dfrac{dV_1}{dt} = -10000 \text{ cm}^3/\text{min}$, $\left.\dfrac{dh}{dt}\right|_{h=2} = 20$

Unknown: $\left.\dfrac{dV_2}{dt}\right|_{h=2}$

$$V = \frac{1}{3}\pi(\frac{h}{3})^2 h = \frac{\pi h^3}{27}$$

$$\frac{dV_{net}}{dt} = \frac{\pi h^2}{9} \cdot \frac{dh}{dt} = \frac{\pi \cdot (200)^2 \cdot 20}{9} = \frac{800000\pi}{9} \approx 279252$$

$$\frac{dV_2}{dt} = \frac{dV_{net}}{dt} - \frac{dV_1}{dt} = 279252 - (-10000) = \boxed{289252 \text{ cm}^3/\text{min}}$$

Water is being pumped into the tank at a rate of 289252 cm^3/min.

Problem 14

Gravel is being dumped from a conveyor belt at a rate of 30 ft^3/min and its coarseness is such that it forms a pile in the shape of a cone whose base diameter and height are always equal. How fast is the height of the pile increasing when the pile is 10 ft hight?

Given: $\dfrac{dV}{dt} = 30 \text{ ft}^3/\text{min}$

Unknown: $\left.\dfrac{dx}{dt}\right|_{x=10}$

$$V = \frac{1}{3}\pi(\frac{x}{2})^2 x = \frac{\pi x^3}{12}$$

$$\frac{dV}{dt} = \frac{\pi x^2}{4} \cdot \frac{dx}{dt}$$

$$\frac{dx}{dt} = \frac{\frac{dV}{dt} \cdot 4}{\pi x^2} = \frac{30 \cdot 4}{\pi \cdot 10^2} = \boxed{\frac{5}{6\pi} \text{ ft/min}}$$

The height of the pile increases at a rate of $\dfrac{5}{6\pi}$ ft/min.

Problem 16

The Boyle's Law states that when a sample of gas is compressed at a constant temperature, the pressure P and the volume V satisfy the equation $PV = C$, where C is a constant. Suppose that at a certain instant the volume is 600 cm^3, the pressures is 150 KPa, and the pressure is increasing at a arate of 20 KPa/min. At what rate is the volume decreasing at this instant?

Given: $\frac{dP}{dt} = 20$ KPa/min

Unknown: $\left.\frac{dV}{dt}\right|_{V=600,P=150}$

$$V = \frac{C}{P}$$

$$\frac{dV}{dt} = C \cdot (-\frac{1}{P^2}) \cdot \frac{dP}{dt} = -\frac{V}{P} \cdot \frac{dP}{dt} = -\frac{600}{150} \cdot 20 = \boxed{-80 \text{ cm}^3/\text{min}}$$

The volume decreases at a rate of -80 cm^3/min.

Problem 17

If two resistors with resistances R_1 and R_2 are connected in parallel, then the total resistance R, measured in ohms (Ω), is given by:

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

If R_1 and R_2 are increasing at a rate of 0.3 Ω/s, respectively, how fast is R changing when $R_1 = 80\Omega$ and $R_2 = 100\Omega$?

Given: $\frac{dR_1}{dt} = \frac{dR_2}{dt} = 0.3\Omega/\text{s}$

Unknown: $\left.\frac{dR}{dt}\right|_{R_1=80\Omega,R_2=100\Omega}$

$$R = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}}$$

$$\frac{dR}{dt} = -\frac{-\frac{1}{R_1^2} \cdot \frac{dR_1}{dt} - \frac{1}{R_2^2} \cdot \frac{dR_2}{dt}}{(\frac{1}{R_1} + \frac{1}{R_2})^2}$$

$$\frac{dR}{dt} = \frac{\frac{0.3}{80^2} + \frac{0.3}{100^2}}{(\frac{1}{80} + \frac{1}{100})^2}$$

$$\boxed{\frac{dR}{dt} = \frac{41}{270}\Omega/\text{s}}$$

R increases at a rate of $\frac{41}{270}\Omega$/s.

Problem 18

Two sides of a triangle have lengths 12 m and 15 m. The angle between them is increasing at a rate of 2°/min. How fast is the length of the third side increasing when the angle between the sides of fixed length is 60°?

Given: $\dfrac{d\theta}{dt} = 2^\circ/\text{min}$, $a = 12$ m, $b = 15$ m

Unknown: $\left.\dfrac{dx}{dt}\right|_{\theta=60^\circ}$

$$x^2 = a^2 + b^2 - 2ab\cos\theta$$
$$x = \sqrt{12^2 + 15^2 - 2\cdot 12\cdot 15\cdot \frac{1}{2}} = 3\sqrt{21}$$
$$2x\cdot\frac{dx}{dt} = 2ab\sin\theta\frac{d\theta}{dt}$$
$$\frac{dx}{dt} = \frac{ab\sin\theta\frac{d\theta}{dt}}{x} = \frac{12\cdot 15\cdot\frac{\sqrt{3}}{2}\cdot\frac{\pi}{90}}{3\sqrt{21}} = \boxed{\frac{\pi}{3\sqrt{7}}\text{ m/min}}$$

The length of the third side increases at a rate of $\dfrac{\pi}{3\sqrt{7}}$ m/min.

Problem 19

The television camera is positioned 4000 ft from the base of a rocket launching pad. Let's assume the rocket rises vertically and its speed is 600 ft/s when it has risen 3000 ft.

(a) How fast is the distance from the television camera to the rocket changing at that moment?

$$D^2 = x^2 + y^2$$
$$2d\cdot\frac{dD}{dt} = 2y\cdot\frac{dy}{dt}$$
$$\frac{dD}{dt} = \frac{2\cdot 3000\cdot 600}{2\cdot 5000}$$
$$\boxed{\frac{dD}{dt} = 360\text{ ft/s}}$$

The distance is increasing at a rate of 360 ft/s.

(b) If the television camera is always kept aimed at the rocket, how fast is the camera's angle of elevation changing at that same moment?

$$\tan\theta = \frac{y}{x}$$
$$\theta = \arctan\frac{y}{x}$$
$$\frac{d\theta}{dt} = \frac{1}{1+(\frac{y}{x})^2}\cdot\frac{dy}{dt}\cdot\frac{1}{x}$$

$$\frac{d\theta}{dt} = \frac{1}{1+(\frac{3}{4})^2} \cdot \frac{600}{4000}$$
$$\boxed{\frac{d\theta}{dt} = \frac{12}{125} \text{ rads/s}}$$

The camera's angle of elevation is increasing at a rate of $\frac{12}{125}$ rads/s.

Problem 20

A plane flies horizontally at an altitude of 5 km and passes directly over a tracking telescope on the ground. When the angle of elevation is $\pi/3$, this angle is decreasing at a rate of $\pi/6$ rad/min. How fast is the plane traveling at that time?

Given: $h = 5$ km, $\frac{d\theta}{dt} = -\frac{\pi}{6}$ rad/min

Unknown: $\frac{dx}{dt}\Big|_{\theta=\frac{\pi}{3}}$

$$\tan\theta = \frac{h}{x} \implies x = \frac{5}{\sqrt{3}}$$
$$\sec^2\theta \cdot \frac{d\theta}{dt} = h \cdot (-\frac{1}{x^2}) \cdot \frac{dx}{dt}$$
$$\frac{dx}{dt} = -\frac{d\theta}{dt} \cdot \frac{x^2\sec^2\theta}{h} = \frac{\pi}{6} \cdot \frac{25}{5\cdot 3} \cdot 4 = \boxed{\frac{10\pi}{9} \text{ km/min}}$$

The plane is traveling at a velocity of $\frac{10\pi}{9}$ km/min.

Problem 21

A plane flying with a constant speed of 300 km/h passes over a ground radar station at an altitude of 1km and climbs at an angle of 30°. At what rate is the distance from the plane to the radar station increasing a minute later?

Given: $v = 300$ km/h, $\theta = \frac{\pi}{6}$

Unknown: $\frac{dD}{dt}\Big|_{t=1}$

$$D^2 = x^2 + y^2$$
$$D^2 = (\frac{\sqrt{3}}{2} \cdot 5t)^2 + (1 + \frac{1}{2} \cdot 5t)^2$$
$$2D \cdot \frac{dD}{dt} = 5\sqrt{3}t \cdot \frac{5\sqrt{3}}{2} + (2+5t) \cdot \frac{5}{2}$$
$$\frac{dD}{dt} = \frac{5\sqrt{3}t \cdot \frac{5\sqrt{3}}{2} + (2+5t) \cdot \frac{5}{2}}{2\sqrt{(\frac{5\sqrt{3}}{2})^2 + (\frac{7}{2})^2}}$$

$$\boxed{\frac{dD}{dt} = \frac{55}{2\sqrt{31}} \text{ km/min}}$$

The distance from the plane to the radar station increases at a rate of $\dfrac{55}{2\sqrt{31}}$ km/min.

Problem 22

A runner sprints around a circular track of radius 100 m at a constant speed of 7 m/sec. The runner's friend is standing at a distance 200 m from the center of the track. How fast is the distance between the friends changing when the distance between them is 200 m?

Given: $r = 100, \dfrac{dx}{dt} = 7$

Unknown: $\left.\dfrac{dD}{dt}\right|_{D=200}$

$$D^2 = x^2 + 200^2 - 2 \cdot 200x\cos\theta$$
$$40000 = 50000 - 40000\cos\theta \implies \cos\theta = \frac{1}{4}$$
$$2D \cdot \frac{dD}{dt} = 400x\sin\theta \cdot \frac{d\theta}{dt}$$
$$\frac{dx}{dt} = 7 \implies \frac{d\theta}{dt} = \frac{7}{100}$$
$$\frac{dD}{dt} = 100 \cdot \frac{7}{100} \cdot \frac{\sqrt{15}}{4}$$
$$\boxed{\frac{dD}{dt} = \frac{7\sqrt{15}}{4} \text{ m/s}}$$

The distance between the friends are changing at a rate of $\dfrac{7\sqrt{15}}{4}$ m/s.